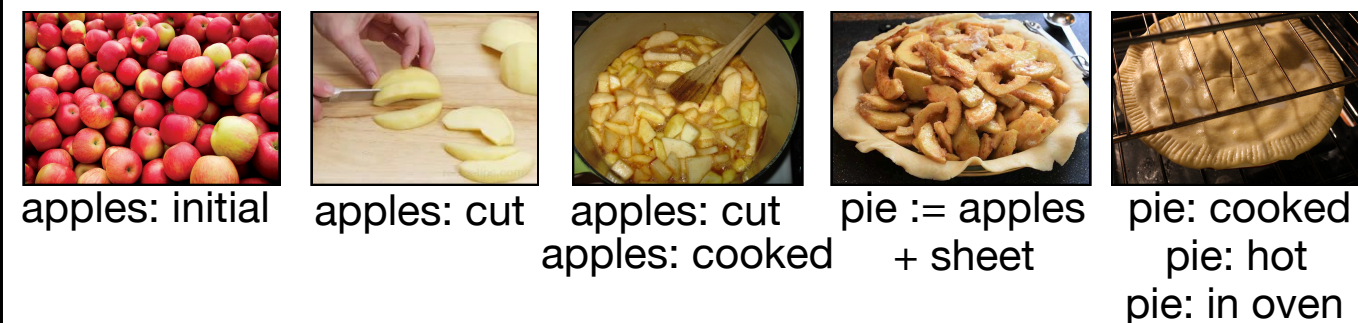


Understanding by Simulation

Understanding by Labeling

- Label syntactic and semantic categories for surface words
- Strong focus on what is explicitly **stated** in text

- 1) Slice apples into pieces.
- 2) Saute them until browned.
- 3) Lay them in the pie sheet.
- 4) Bake for 30 minutes.

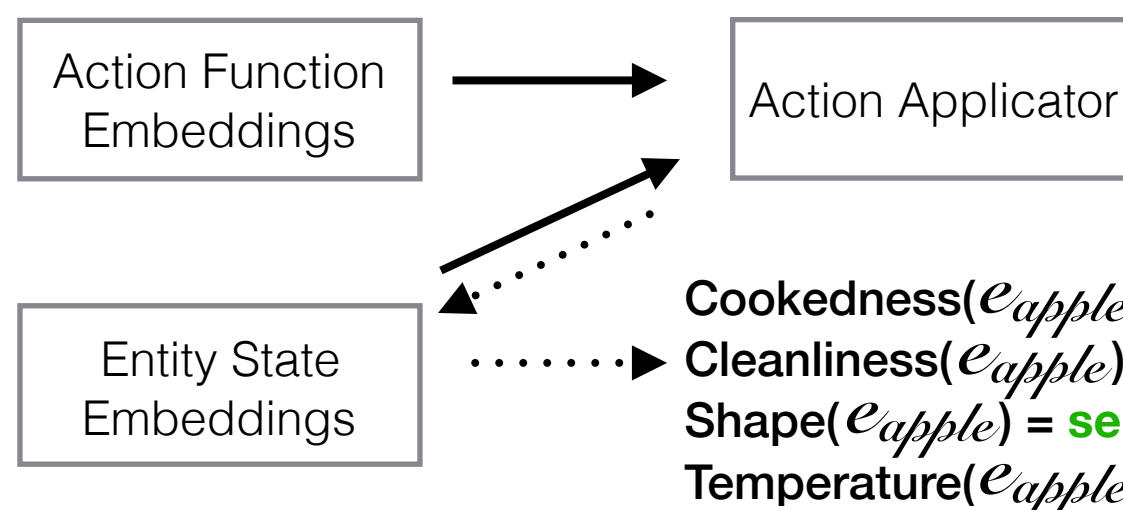


Understanding by Simulation

- Simulation of causal effects of descriptions in text
- Strong focus on what is **implied** in text
- MemNNs (Sukhbataar et al., 2015); EntNet (Henaff et al., 2016)

Action-Oriented Simulation

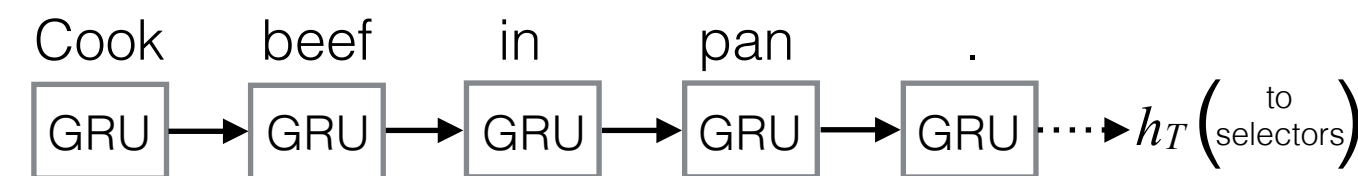
“Wash and slice the apples”



Approach:

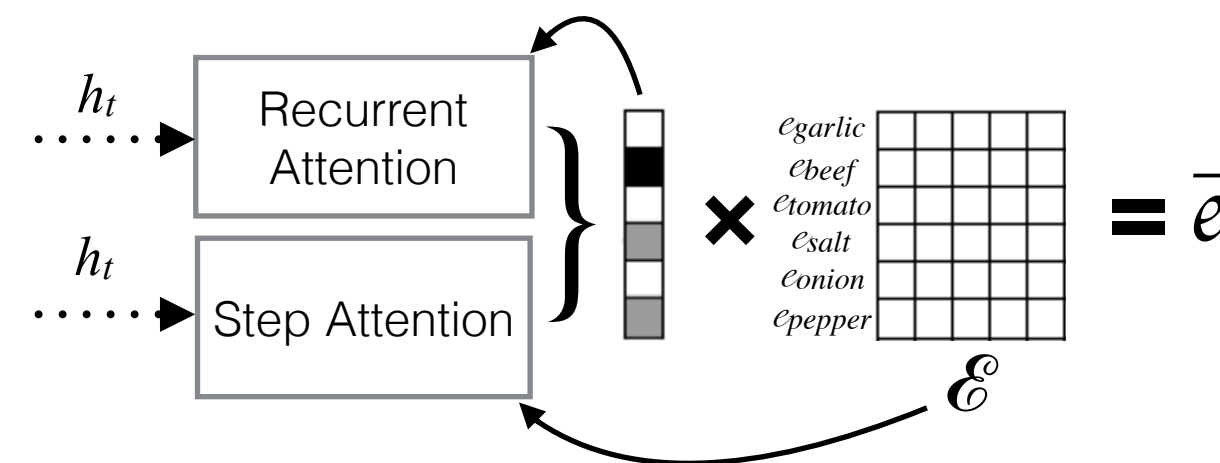
- Action function embeddings **transform** entity state embeddings
- Entity states **tracked** through time by embedding representation
- Current state of entity **encoded** in state embedding
- Crowdsource end state labels induced by actions
- Errors in predicting state backpropagate to action functions

Sentence Encoder



(1) Encode sentence

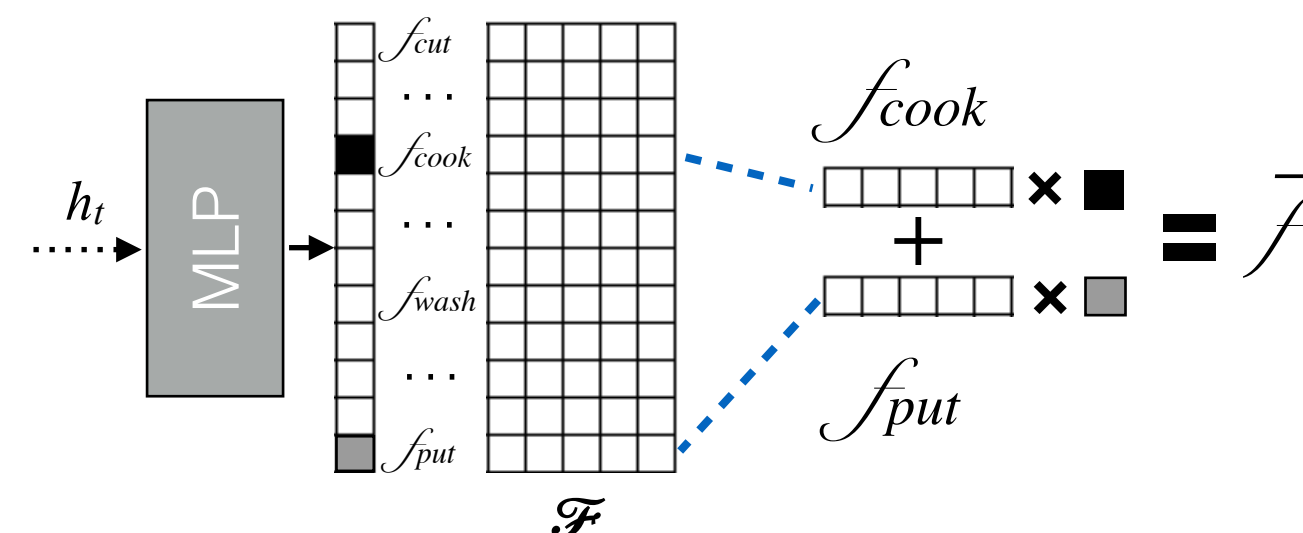
Entity Selector



(2a) Select Entities

- Step Attention:** Compute individual attention weight for each entity embedding
- Recurrent Attention:** Attend over step's attention weights and previous step's attention
- Use combined attention to compute entity vector

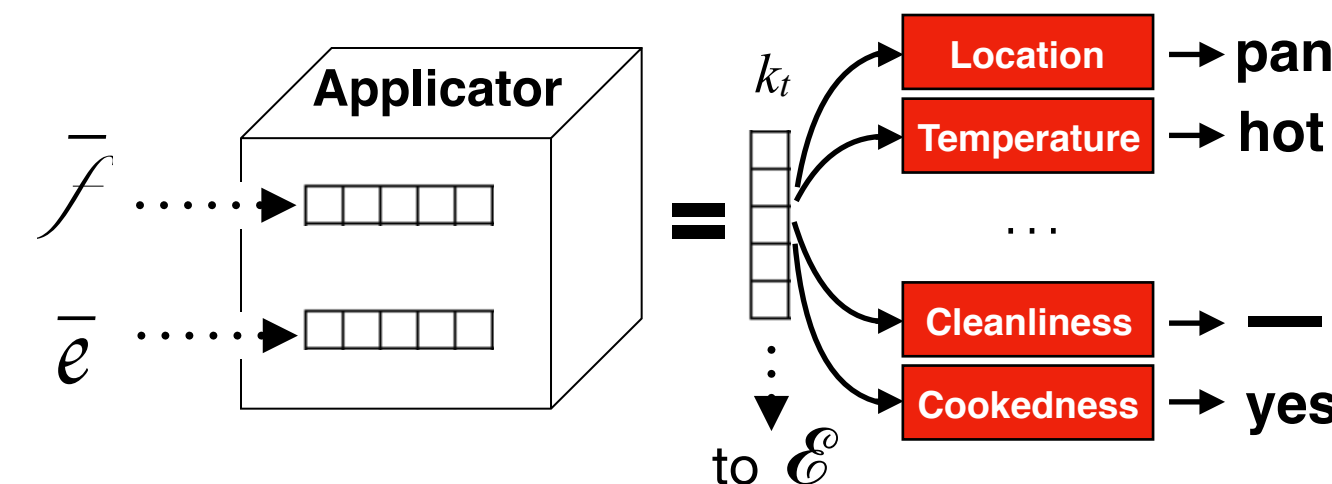
Action Selector



(2b) Select Actions

- Compute unnormalized attention weights over action embeddings
- Compute action vector by weighing action embeddings by their attention weight

Simulation Module



(3) Apply action to entity

- Compute bilinear projection of action and entity vectors

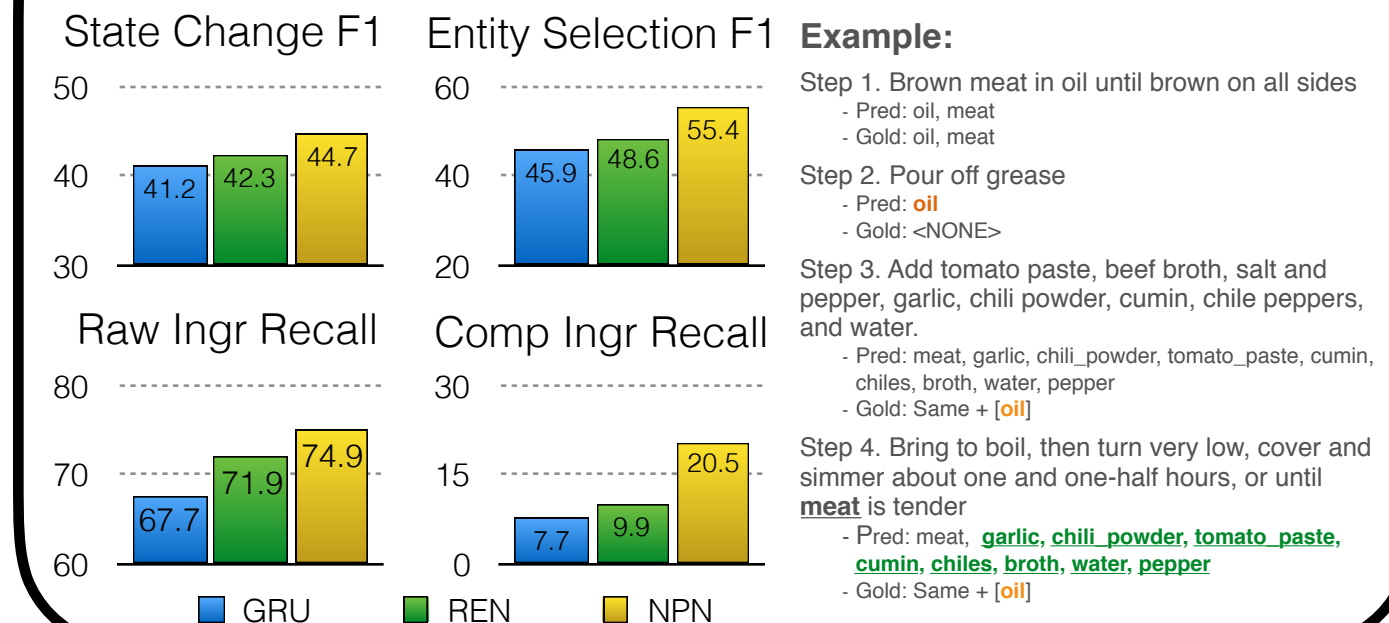
(4) Predict state changes

- Individual classifiers predict state changes of new entity

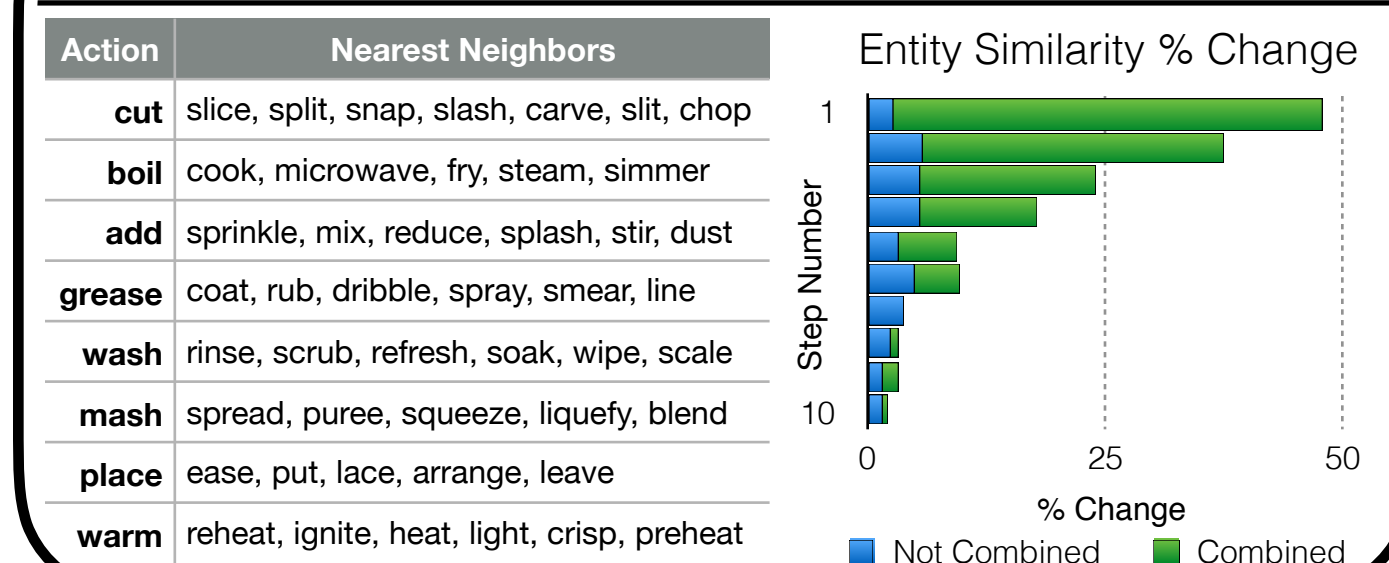
(5) Update Entity Memory

- Entity embeddings updated relative to attention weight during selection (2a)

Tracking



Simulating



Generating

